

MA 301
Test 2
Spring - 2014

Problem	points	out of
1		8
2		8
3		18
4		16
Score		50

Name: SOLUTIONS

HR: _____

Signature: _____

READ BEFORE YOU START

- * Show all work.
- * Change your calculator mode to RADIANS.
- * Sloppy solutions may be penalized.
- * Use calculators ONLY for computations.
- * You must write up every problem in a clear and organized way.
- * If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Suppose $F = \left[\cos(\pi x), xz, \frac{1}{2}x^2 + \ln y \right]$ and S the surface: $z = 4 - y^2$ where $y \geq 0$ and $0 \leq x \leq 3$ with outward pointing normal (away from the x -axis). Let C be the curve which bounds S .

Compute $\oint_C F \cdot dr = \iint_S \text{curl } \vec{F} \cdot \vec{n} dS = \iint_R \text{curl } \vec{F}(\vec{r}) \cdot \vec{N} du dv = *$

$$\text{curl } \vec{F} = \left[\frac{1}{y} - e^x, -x, ze^x \right]$$

$$\vec{r}(u, v) = [u, v, 4 - v^2]$$

$$0 \leq u \leq 3$$

$$0 \leq v \leq 2$$

$$\vec{r}_u = [1, 0, 0]$$

$$\vec{r}_v = [0, 1, -2v]$$

$$\vec{N} = [0, 2v, 1]$$

$$\text{curl } \vec{F}(\vec{r}) = \left[\frac{1}{v} - u, -u, 4 - v^2 \right]$$

$$* = \int_0^2 \int_0^3 (-2uv + 4 - v^2) du dv$$

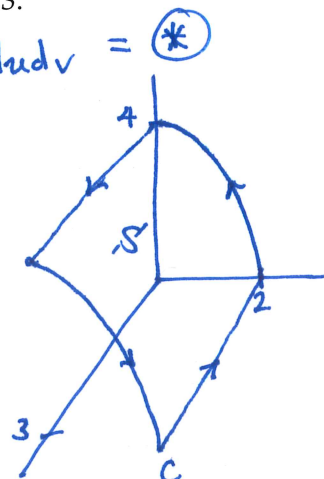
$$= \int_0^2 \left. -u^2 v + 4u - uv^2 \right|_0^3 dv$$

$$= \int_0^2 (-9v + 12 - 3v^2) dv$$

$$= \left. -\frac{9}{2}v^2 + 12v - v^3 \right|_0^2$$

$$= -18 + 24 - 8$$

$$= -2$$



Problem 2: Let $\mathbf{F} = [107, ze^{xz}, xyz]$ and S be the surface (with outward orientation) of the solid bounded by the surfaces $x = 0$, $x = 5$, $y = 0$, $z = 0$, and $y^2 + z^2 = 1$ in the first octant.

Compute $\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_T \operatorname{div} \vec{\mathbf{F}} dV = *$

$$\operatorname{div} \vec{\mathbf{F}} = xy$$

$$* = \iiint_T xy dV$$

$$= \int_0^5 \int_0^{\pi/2} \int_0^1 x r \cos \theta r dr d\theta dx$$

$$= \left(\int_0^5 x dx \right) \left(\int_0^{\pi/2} \cos \theta d\theta \right) \left(\int_0^1 r^2 dr \right)$$

$$= \left(\frac{x^2}{2} \Big|_0^5 \right) \left(\sin \theta \Big|_0^{\pi/2} \right) \left(\frac{r^3}{3} \Big|_0^1 \right)$$

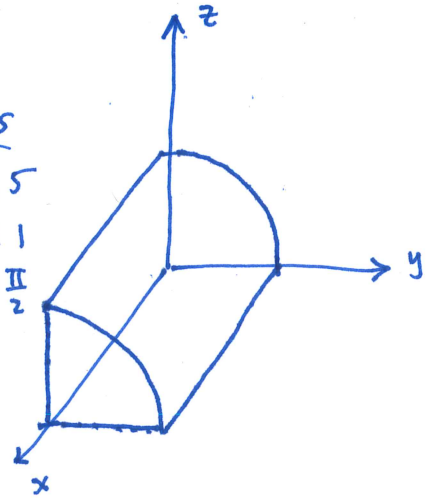
$$= \left(\frac{25}{2} \right) (1) \left(\frac{1}{3} \right) = \frac{25}{6}$$

Cylindrical Coordinates

$$x = x \quad 0 \leq x \leq 5$$

$$y = r \cos \theta \quad 0 \leq r \leq 1$$

$$z = r \sin \theta \quad 0 \leq \theta \leq \frac{\pi}{2}$$



Problem 3: Let $z_1 = -3i$, $z_2 = 3 - 3i$, $z_3 = -5$, $z_4 = -\sqrt{3} - i$.

(a) Perform the following operations and leave answer in standard form. (No Decimals)

$$\triangleright z_1 - z_4 = -3i - (-\sqrt{3} - i) = \sqrt{3} - 2i$$

$$\triangleright \frac{z_4}{z_2} = \frac{(-\sqrt{3} - i)(3 - 3i)}{(3 + 3i)(3 - 3i)} = \frac{-3\sqrt{3} - 3 + 3\sqrt{3}i - 3i}{9 + 9} = \frac{-3\sqrt{3} - 3}{18} + \left(\frac{3\sqrt{3} - 3}{18}\right)i$$

(b) Write z_1, z_2, z_3, z_4 in Euler's polar form. (No Decimals)

$$\begin{aligned} r=3 \quad \theta=-\frac{\pi}{2} &\triangleright z_1 = 3e^{i(-\frac{\pi}{2})} \\ r=\sqrt{18} \quad \theta=-\frac{\pi}{4} &\triangleright z_2 = \sqrt{18}e^{i(-\frac{\pi}{4})} \\ r=5 \quad \theta=\pi &\triangleright z_3 = 5e^{i\pi} \\ r=2 \quad \theta=\frac{7\pi}{6} &\triangleright z_4 = 2e^{i(\frac{7\pi}{6})} \end{aligned}$$

(b) Perform the following operations and leave answer in Euler's polar form. (No Decimals)

$$\triangleright z_1 z_2 = (3e^{i(-\frac{\pi}{2})})(\sqrt{18}e^{i(-\frac{\pi}{4})}) = 3\sqrt{18}e^{i(-\frac{3\pi}{4})}$$

$$\triangleright (z_1 z_2)^6 = (3\sqrt{18})^6 e^{i(-\frac{18\pi}{4})}$$

(d) Solve the following complex equations for x and leave your solutions in Euler's Form: $re^{i\theta}$. (No Decimals)

$$\triangleright \frac{x^{1/6}}{z_1} - z_2 = 0 \Rightarrow x = (z_1 z_2)^6 = (3\sqrt{18})^6 e^{i(-\frac{18\pi}{4})}$$

$$\triangleright x^3 - z_3 = 0 \Rightarrow x = (5e^{i\pi})^{1/3} = 5^{1/3} e^{i\frac{\pi}{3} + \frac{2\pi}{3}k}$$

$$k=0, 1, 2$$

$$k=0: x = 5^{1/3} e^{i\frac{\pi}{3}}$$

$$k=1: x = 5^{1/3} e^{i\pi}$$

$$k=2: x = 5^{1/3} e^{i\frac{5\pi}{3}}$$

Problem 4: Answer the following quick computation questions:

(a) Let $\overset{M}{F} = \overset{N}{yz}\mathbf{i} + \overset{P}{xz}\mathbf{j} + xy\mathbf{k}$ and S be a square of side length 7 inside the plane $x + y + z = 10$. If C is the boundary of this square (oriented counterclockwise) then find $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot \vec{n} \, dS$

$$\text{curl } \vec{F} = [x-x, y-y, z-z]$$

$$= [0, 0, 0]$$

$$\Rightarrow \oint_C \vec{F} \cdot d\mathbf{r} = 0$$

(b) Let $\overset{M}{F} = -y\mathbf{i} + \overset{N}{x}\mathbf{j} + \overset{P}{e^z}\mathbf{k}$ and S be a circle of radius 2 inside the plane $ax + by + cz = 10$. If C is the boundary of this circle (oriented counterclockwise) then find $\oint_C \mathbf{F} \cdot d\mathbf{r}$ (your answer should be in terms of $a, b, & c$)

$$\text{curl } \vec{F} = [0-0, 0-0, 1+1]$$

$$= [0, 0, 2]$$

$$\text{normal vector of } S = [a, b, c]$$

$$\Rightarrow \text{unit normal of } S = \frac{1}{\sqrt{a^2+b^2+c^2}} [a, b, c]$$

$$\iint_S \text{curl } \vec{F} \cdot \vec{n} \, dS = \iint_S \frac{2c}{\sqrt{a^2+b^2+c^2}} \, dS$$

$$= \frac{2c}{\sqrt{a^2+b^2+c^2}} \text{Area}(S)$$

$$= \frac{8\pi c}{\sqrt{a^2+b^2+c^2}}$$

(c) Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where $\overset{M}{F} = \overset{N}{5y}\mathbf{i} + \overset{P}{2z}\mathbf{j} + 5x\mathbf{k}$ and C is the boundary around the triangle obtained by tracing out the path $(2, 0, 0)$ to $(2, 0, 2)$, to $(2, 2, 2)$ back to $(2, 0, 0)$.

$$\text{curl } \vec{F} = [0-2, 0-5, 0-5]$$

$$= [-2, -5, -5]$$



$$\mathbf{n} = [-1, 0, 0]$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S 2 \, dS = 2 \text{Area}(S) = 4$$

(d) Compute $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$ where S is the surface of the solid cube with $0 \leq x \leq 2$, $|y| \leq 1$, and $|z| \leq 1$ with outward pointing normal.

$$\text{div } \vec{F} = 10x - 3x - 3 - 7x = -3$$

$$\iint_S \vec{F} \cdot \vec{n} \, dS = \iiint_T -3 \, dV = -3 \text{Vol}(T)$$

$$= -3(2)^3 = -24$$